National University of Computer and Emerging Sciences



Laboratory Manuals

for

Computer Networks - Lab

(CL -3001)

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Lab Manual 03

Objective:

- Analyzing the ICMP packets using Wireshark
- Analyzing the FTP packets using Wireshark

Instructions:

- Submit one PDF/Word document containing answers and relevant screenshots.
- File name: RollNo_Lab03
- Submit through LMS before the announced deadline.
- Screenshots must clearly show the relevant Wireshark packet/filter or terminal output.

Lab Statement 1: Capturing FTP packets using Wireshark (10)

Step 1: **Start a Wireshark capture.**

a. Close all unnecessary network traffic, such as the web browser, to limit the amount traffic during the Wireshark capture.

b. Start the Wireshark capture.

Step 2: Open the provided FTP session capture.

Open the provided FTP_Session.pcapng file in Wireshark. The capture contains an FTP session in which a client connects to an FTP server and downloads the file legal.txt.

Carefully analyze the packets in Wireshark windows and answer the following question:

Use the FTP_Session.pcapng (Wireshark Capture File) to answer the questions below

1. FTP uses a control connection and a separate data connection. Apply the filters tcp.port == 20 and tcp.port == 21. Identify the purpose of port 21 and explain the role of port 20 in the FTP session. Also identify the port numbers used for the data connection in the provided trace, if different.

2. Filter out each packet using either FTP or FTP-DATA Protocol (using **ftp || ftp-data** filter). For each FTP/FTP-DATA packet, record the packet number, FTP response code (where applicable), response argument, and briefly explain the purpose of the packet in the FTP session.

Lab Statement 2: Analyzing ICMP Packets using Wireshark (10)

- **Step 1:** Run Wireshark
- **Step 2:** Load the Session file **ICMP_Session**

- **Step 3:** Now filter out all non-ICMP packets by typing "icmp" (without quotes) in the filter field towards the top of the Wireshark window

- **Step 4:** Analyze the ICMP Packets and answer the following questions

1- Are ICMP messages sent over UDP or TCP?	
2- Identify the source host's link-layer (e.g., Ethernet) MAC address from the captured ICMP packet.	
3- What type of ICMP request is being sent by the host? Identify the ICMP message type.	
4- How many ICMP Echo Request messages are sent by the source host in the captured session?	
5- What are the source IP address and destination IP address in the ICMP Echo Request packet?	
6- Why does an ICMP packet not contain source and destination port numbers like TCP and UDP packets?	
7- Examine an ICMP Echo Request and its corresponding Echo Reply. Which ICMP header fields distinguish the request from the reply? State the values of these fields for both packets.	
8- Examine one ICMP Echo Request packet. What are its ICMP Type and Code values? Identify the other fields present in the ICMP header. Also report the size of the Checksum, Identifier, and Sequence Number fields.	

9- Examine the corresponding ICMP Echo Reply packet. What are its ICMP Type and Code values? Identify the other fields present in the ICMP header. Also report the size of the Checksum, Identifier, and Sequence Number fields.	
10- Examine packet no. 56. What are its ICMP Type and Code values? Why does the ICMP message contain the original IP and TCP headers?	